

Oxidation - Reduction

- What is the oxidizing agent in the following reaction?
 $\text{Fe}(\text{CN})_6^{3-} + \text{Co}(\text{CN})_5^{3-} + \text{CN}^- \rightarrow \text{Fe}(\text{CN})_6^{4-} + \text{Co}(\text{CN})_6^{3-}$
 (A) $\text{Fe}(\text{CN})_6^{3-}$ (B) $\text{Co}(\text{CN})_5^{3-}$
 (C) CN^- (D) $\text{Co}(\text{CN})_6^{3-}$
- What are the oxidation states of N in NH_4NO_3 ?
 (A) Both are -3
 (B) Both are $+1$
 (C) One is -1 and one is $+1$
 (D) One is -3 and one is $+5$
- A 0.4679 g sample of iron-containing ore is dissolved in acid and the iron reduced to the $+2$ oxidation state. The solution is titrated with 0.03930 M KMnO_4 , requiring 26.82 mL to reach the equivalence point. What is the percentage by mass of iron in the ore?
 (A) 12.58% (B) 29.43%
 (C) 62.91% (D) 72.36%
- Ships sometimes have strips of titanium welded to their steel hulls to protect them from corrosion. Which is the best explanation for this protective action?
 (A) Ti is more easily oxidized than Fe.
 (B) Ti and Fe form an alloy that stabilizes the Fe.
 (C) Ti^{2+} has a more positive reduction potential than Fe^{2+} .
 (D) The Ti and Fe form a galvanic cell with Ti being reduced.
- Hypophosphorous acid (H_3PO_2) disproportionates in acidic solution to give phosphine (PH_3) and phosphorous acid (H_3PO_3). What ratio of $\text{PH}_3 : \text{H}_3\text{PO}_3$ is produced?
 (A) 1 : 1 (B) 1 : 2 (C) 2 : 3 (D) 3 : 2
- A solution containing 0.0100 mol of silver nitrate and 0.0100 mol copper(II) nitrate is electrolyzed with a current of 0.170 A for 152 minutes. What mass of metal is deposited on the cathode?
 (A) 1.12 g (B) 1.27 g
 (C) 1.38 g (D) 1.73 g
- What is the average oxidation state of iron in halotrichite, $\text{FeAl}_2(\text{SO}_4)_4 \cdot 22 \text{H}_2\text{O}$?
 (A) 0 (B) $+2$ (C) $+2.5$ (D) $+3$
- Which half-reaction takes place at the anode during electrolysis of aqueous sodium iodide?
 (A) $\text{Na}^+(\text{aq}) + e^- \rightarrow \text{Na}(\text{s})$
 (B) $2 \text{H}_2\text{O}(\text{l}) + 2 e^- \rightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
 (C) $2 \text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{s}) + 2 e^-$
 (D) $2 \text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^-$
- In an electrochemical cell, the reaction shown below occurs. Which changes will decrease the cell potential?

$$\text{Fe}^{2+}(\text{aq}) + \text{Ag}^+(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$$
 I. Decreasing the concentration of $\text{Fe}^{2+}(\text{aq})$
 II. Increasing the size of the silver electrode
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
- 180 mL of a solution that is 0.024 M in both AgNO_3 and $\text{Cu}(\text{NO}_3)_2$ is electrolyzed with a current of 0.70 A for 350 s. What deposits on the cathode?
 (A) Ag only
 (B) Cu only
 (C) 2.5×10^{-3} mol Ag and some Cu
 (D) 2.5×10^{-3} mol Cu and some Ag
- What is the standard potential of a fuel cell where methanol is oxidized to carbon dioxide and oxygen is reduced to water?

Half-reaction	E° , V
$\text{CO}_2(\text{g}) + 6 \text{H}^+(\text{aq}) + 6 e^- \rightarrow \text{CH}_3\text{OH}(\text{l}) + \text{H}_2\text{O}(\text{l})$	-0.02
$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^- \rightarrow 2 \text{H}_2\text{O}(\text{l})$	1.23

 (A) 0.30 V (B) 0.31 V
 (C) 1.21 V (D) 1.25 V
- The half-cell potential at 298 K for the reduction of an oxoanion becomes more negative by 0.118 V for each increase of 1 pH unit. How many protons and electrons are involved in this half-reaction?
 (A) 1 proton, 1 electron (B) 1 proton, 2 electrons
 (C) 2 protons, 1 electron (D) 3 protons, 2 electrons
- What is the oxidation number of carbon in methanol, CH_3OH ?
 (A) -2 (B) 0 (C) $+2$ (D) $+4$

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14. A standard galvanic cell is assembled from silver and platinum in 1 M solutions of the corresponding metal ions. Which metal would be labeled the cathode, and what is the voltage of the standard galvanic cell?

Half-reaction	E° , V
$\text{Pt}^{2+}(aq) + 2e^- \rightarrow \text{Pt}(s)$	1.20
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	0.80

- (A) Ag cathode, $E^\circ = 1.60$ V
 (B) Pt cathode, $E^\circ = 1.20$ V
 (C) Ag cathode, $E^\circ = 0.80$ V
 (D) Pt cathode, $E^\circ = 0.40$ V

15. A spoon is electroplated in a solution of 1.0 M silver nitrate with a current of 0.10 A. How long does it take to plate 0.10 g of silver on the spoon?

- (A) 5.0 min (B) 10. min
 (C) 15. min (D) 30. min

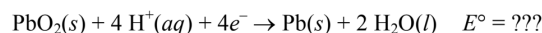
16. Aqueous sulfuric acid containing 1.0 M NaI is electrolyzed at glassy carbon electrodes. What is produced at the anode?

- (A) H_2 (B) O_2
 (C) I_2 (D) SO_2

17. A certain galvanic cell has a standard cell potential that increases with increasing temperature. Which best explains this observation?

- (A) $\Delta H^\circ_{\text{rxn}} > 0$ (B) $\Delta S^\circ_{\text{rxn}} > 0$
 (C) $\Delta G^\circ_{\text{rxn}} > 0$ (D) $E^\circ > 0$

18. What is the standard reduction potential under acidic conditions for the reduction of $\text{PbO}_2(s)$ to $\text{Pb}(s)$?



Half-reaction	E° , V
$\text{PbO}_2(s) + 4 \text{H}^+(aq) + 2e^- \rightarrow \text{Pb}^{2+}(aq) + 2 \text{H}_2\text{O}(l)$	1.468
$\text{Pb}^{2+}(aq) + 2e^- \rightarrow \text{Pb}(s)$	-0.126

- (A) 0.671 V (B) 0.797 V
 (C) 1.342 V (D) 1.594 V

19. What is the average oxidation state of sulfur in calcium tetrathionate, CaS_4O_6 ?

- (A) -2.0 (B) +2.0 (C) +2.5 (D) +3.0

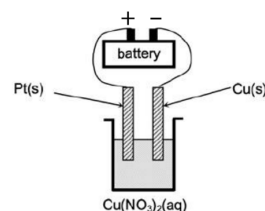
20. When the equation for the oxidation of $\text{Cu}(s)$ by $\text{NO}_3^-(aq)$ to give $\text{Cu}^{2+}(aq)$ and $\text{NO}(g)$ in acidic aqueous solution is balanced using smallest whole number coefficients, what is the coefficient for $\text{NO}_3^-(aq)$?

- (A) 1 (B) 2 (C) 3 (D) 6

21. Water is electrolyzed with a current of 10.0 A until 132 L of hydrogen gas (after drying, measured at STP) has been produced. How much time does this take?

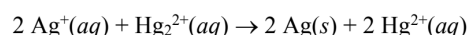
- (A) 7.90 h (B) 15.8 h (C) 31.6 h (D) 63.2 h

22. A 1.0 M solution of $\text{Cu}(\text{NO}_3)_2$ is electrolyzed using the setup shown below. What reaction occurs at the cathode?



- (A) $\text{Pt}^{2+}(aq) + 2e^- \rightarrow \text{Pt}(s)$
 (B) $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$
 (C) $\text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4e^- \rightarrow 4 \text{OH}^-(aq)$
 (D) $2 \text{H}_2\text{O}(l) + 2e^- \rightarrow \text{H}_2(g) + 2 \text{OH}^-(aq)$

23. What is E° for the oxidation of mercury(I) ion by silver ion?



Half-reaction	E° , V
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$2 \text{Hg}^{2+}(aq) + 2e^- \rightarrow \text{Hg}_2^{2+}(aq)$	+0.91

- (A) -0.11 V (B) 0.69 V
 (C) 1.71 V (D) 2.51 V

24. What is the standard reduction potential of $\text{PbF}_2(s)$ at 298 K? The K_{sp} of PbF_2 at 298 K is 4.0×10^{-8} .

Half-reaction	E° , V
$\text{Pb}^{2+}(aq) + 2e^- \rightarrow \text{Pb}(s)$	-0.13
$\text{PbF}_2(s) + 2e^- \rightarrow \text{Pb}(s) + 2 \text{F}^-(aq)$	???

- (A) -0.35 V (B) -0.22 V
 (C) -0.13 V (D) -0.04 V

25. What is the average oxidation state of iron in riebeckite, $\text{Na}_2\text{Fe}_5\text{Si}_8\text{O}_{22}(\text{OH})_2$?

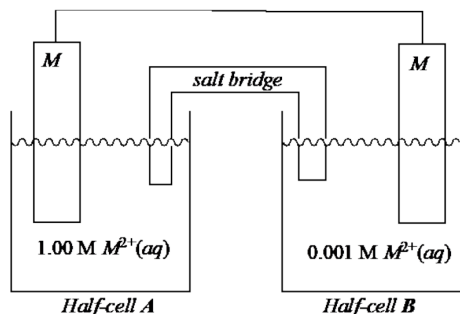
- (A) +2.0 (B) +2.4 (C) +3.0 (D) +5.6

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26. An electrolytic cell is operated for 3000 s using a current of 1.50 A. From which 1.0 M solution will the smallest mass of metal be deposited?

(A) CdCl_2 (B) InBr_3
(C) TlNO_3 (D) $\text{Pb(NO}_3)_2$

27. A galvanic cell using an unknown metal M is arranged as shown below. What can be concluded about this cell?



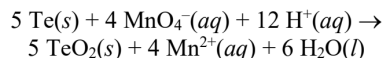
- (A) Half-cell A is the cathode.
(B) Half-cell B is the cathode.
(C) No current flows in the cell since the same metal is used in both half-cells.
(D) Which cell is the cathode depends on the standard reduction potential of $\text{M}^{2+}(\text{aq})$.

28. When manganese metal is placed in a solution containing silver(I) ion in acidic solution, metallic silver deposits on the manganese. Which half-reactions are taking place?

I. $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$
II. $\text{Mn}(\text{s}) + 4 \text{H}_2\text{O}(\text{l}) \rightarrow \text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 7 \text{e}^-$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

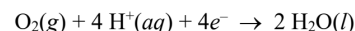
29. What is E° for the following redox reaction?



Half-reaction	E°, V
$\text{TeO}_2(\text{s}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \rightarrow \text{Te}(\text{s}) + 2 \text{H}_2\text{O}(\text{l})$	+0.59
$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$	+1.51

- (A) -3.09 V (B) 0.92 V
(C) 3.09 V (D) 5.19 V

30. The reduction of O_2 to H_2O in acidic solution has a standard reduction potential of +1.23 V. What is the effect on the half-cell potential at 25 °C when the pH of the solution is decreased by one unit?

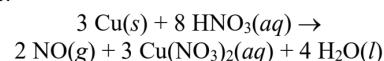


- (A) The half-cell potential decreases by 236 mV.
(B) The half-cell potential increases by 236 mV.
(C) The half-cell potential decreases by 59 mV.
(D) The half-cell potential increases by 59 mV.

31. Romanèchite has the formula $\text{Ba}_2\text{Mn}_5\text{O}_{10} \cdot 2\text{H}_2\text{O}$. What is the average oxidation state of manganese in romanèchite?

(A) +2.0 (B) +3.2 (C) +3.6 (D) +4.0

32. Which element is being reduced in the following equation?



(A) Cu (B) H (C) N (D) O

33. A galvanic cell is constructed under standard conditions using cobalt in cobalt(II) nitrate solution and indium in indium(III) nitrate solution. Which statements about this cell are correct?

Half-reaction	E°, V
$\text{Co}^{2+}(\text{aq}) + 2 \text{e}^- \rightarrow \text{Co}(\text{s})$	-0.28
$\text{In}^{3+}(\text{aq}) + 3 \text{e}^- \rightarrow \text{In}(\text{s})$	-0.34

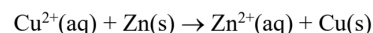
I. The standard cell potential is 0.46 V.
II. Indium is the anode.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

34. Molten sodium chloride is electrolyzed for 10.0 minutes with a constant current of 10.0 A. What mass of sodium metal is produced?

(A) 1.43 g (B) 2.20 g (C) 3.63 g (D) 4.41 g

35. The standard cell potential of a galvanic cell based on the reaction below is 1.10 V. Which concentrations would result in the largest measured voltage?



- (A) $[\text{Cu}^{2+}] = 1.0 \text{ M}$ and $[\text{Zn}^{2+}] = 1.0 \text{ M}$
(B) $[\text{Cu}^{2+}] = 3.0 \text{ M}$ and $[\text{Zn}^{2+}] = 1.0 \text{ M}$
(C) $[\text{Cu}^{2+}] = 1.0 \text{ M}$ and $[\text{Zn}^{2+}] = 3.0 \text{ M}$
(D) $[\text{Cu}^{2+}] = 3.0 \text{ M}$ and $[\text{Zn}^{2+}] = 3.0 \text{ M}$

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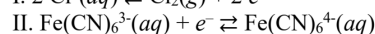
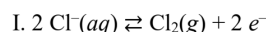
36. The standard reduction potential of oxygen under acidic conditions at 298 K is +1.23 V. What is the standard reduction potential for the four-electron reduction of $O_2(g)$ under basic conditions?

(A) 1.23 V (B) 1.02 V (C) 0.83 V (D) 0.40 V

37. What is the average oxidation state of tin in the mineral abhurite, $Sn_{21}Cl_{16}(OH)_{14}O_6$?

(A) +1.71 (B) +2.00 (C) +2.76 (D) +3.43

38. If used in an electrochemical cell, which of the following half-reactions would require an inert electrode?



(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

39. Chromium is electroplated industrially by the electrolysis of solutions of $K_2Cr_2O_7$. How much time would be required to deposit 1.00 kg of Cr using a current of 200.0 A?

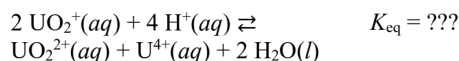
(A) 2.58 h (B) 7.74 h (C) 15.5 h (D) 31.0 h

40. What is the standard reduction potential of $Cr^{3+}(aq)$ to form metallic chromium?

Half-reaction	E° , V
$Cr^{3+}(aq) + e^- \rightarrow Cr^{2+}(aq)$	-0.41
$Cr^{2+}(aq) + 2e^- \rightarrow Cr(s)$	-0.91
$Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$	???

(A) -0.74 V (B) -1.32 V
(C) -1.73 V (D) -2.23 V

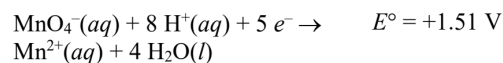
41. What is the equilibrium constant at 298 K for the disproportionation of uranium(V) in acidic solution as shown below?



Half-reaction	E° , V
$UO_2^{2+}(aq) + e^- \rightarrow UO_2^+(aq)$	+0.16
$UO_2^+(aq) + 4 H^+(aq) + e^- \rightarrow U^{4+}(aq) + 2 H_2O(l)$	+0.27

(A) 1.9×10^{-4} (B) 73
(C) 5300 (D) 1.9×10^7

42. The standard reduction potential E° for the reduction of permanganate in acidic solution is +1.51 V. What is the reduction potential for this half-reaction at pH = 5.00?

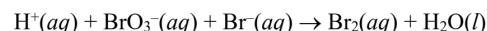


(A) +1.50 V (B) +1.42 V
(C) +1.04 V (D) -0.85 V

43. What is the average oxidation state of tungsten in sodium phosphotungstate, $Na_3PW_{12}O_{40}$?

(A) +6.00 (B) +6.17
(C) +6.42 (D) +6.67

44. Bromate ion and bromide ion react to form bromine in acidic solution. When the reaction is balanced, which statement about $H^+(aq)$ is correct?



(A) Its coefficient is twice the coefficient of $Br^-(aq)$.
(B) Its coefficient is twice the coefficient of $Br_2(aq)$.
(C) Its coefficient is twice the coefficient of $BrO_3^-(aq)$.
(D) Its coefficient is the sum of the coefficients of $BrO_3^-(aq)$ and $H_2O(l)$.

45. Which reactions are spontaneous under standard conditions?

Half-Reaction	E° , V
$Cu^+(aq) + e^- \rightarrow Cu(s)$	+0.52
$Co^{2+}(aq) + 2 e^- \rightarrow Co(s)$	-0.28
$In^{3+}(aq) + 3 e^- \rightarrow In(s)$	-0.34

- I. $2 Cu^+(aq) + Co(s) \rightarrow Co^{2+}(aq) + 2 Cu(s)$
II. $3 Co^{2+}(aq) + 2 In(s) \rightarrow 2 In^{3+}(aq) + 3 Co(s)$

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

46. An electrolytic cell consists of two copper electrodes immersed in a solution of copper(II) sulfate. What is the result of passing 0.35 A of current through this cell for 1300 s?

(A) The mass of the anode increases by 0.15 g.
(B) The mass of the anode increases by 0.30 g.
(C) The mass of the anode decreases by 0.15 g.
(D) The mass of the anode decreases by 0.30 g.

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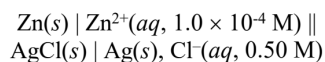
47. What is the change in standard free energy at 298 K for the conversion of ozone to molecular oxygen as shown in the equation below?



Half-Reaction	E°, V
$\text{O}_3(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 e^- \rightarrow \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	+2.08
$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^- \rightarrow 2 \text{H}_2\text{O}(\text{l})$	+1.23

- (A) -164 kJ mol^{-1} (B) -328 kJ mol^{-1}
 (C) -401 kJ mol^{-1} (D) -492 kJ mol^{-1}

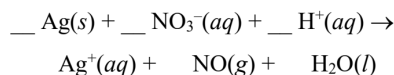
48. What is the cell potential of the following galvanic cell under the stated conditions at 298 K?



Half-Reaction	E°, V
$\text{Zn}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Zn}(\text{s})$	-0.762
$\text{AgCl}(\text{s}) + e^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.222

- (A) +0.884 V (B) +0.984 V
 (C) +1.084 V (D) +1.120 V

49. When the chemical equation is balanced, what is the ratio of the coefficient of Ag(s) to the coefficient of H₂O(l)?



- (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 3 : 2

50. A current of 0.44 A is passed through a solution of a ruthenium nitrate salt, causing reduction of the metal ion to the metal. After 25.0 minutes, 0.345 g of Ru(s) has been deposited. What is the oxidation state of ruthenium in the nitrate salt?

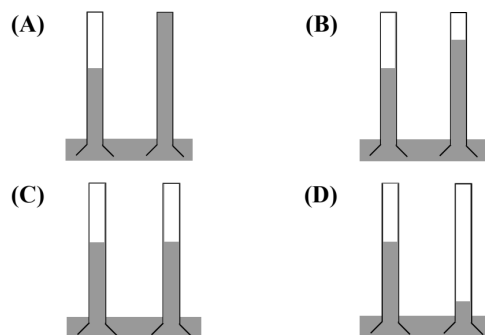
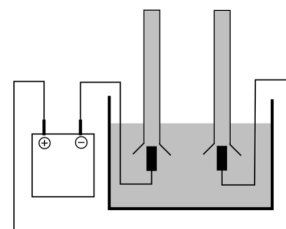
- (A) +2 (B) +3 (C) +4 (D) +6

51. What is the K_{sp} of Ca(OH)₂ at 298 K?

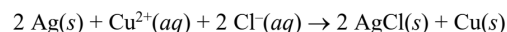
Half-Reaction	E°, V
$\text{Ca}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Ca}(\text{s})$	-2.87 V
$\text{Ca}(\text{OH})_2(\text{s}) + 2 e^- \rightarrow \text{Ca}(\text{s}) + 2 \text{OH}^-(\text{aq})$	-3.02 V

- (A) 1.6×10^{-6} (B) 3.1×10^{-6}
 (C) 8.4×10^{-6} (D) 2.9×10^{-3}

52. Electrolysis of water containing sulfuric acid as an electrolyte is carried out as shown and the gases produced at the electrodes collected in two identical tubes, both initially filled with the sulfuric acid solution. After electrolysis has been carried out for a certain time, which picture best represents the appearance of the tubes?



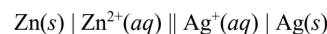
53. What is ΔE° for this reaction?



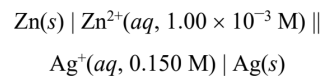
Half-Reaction	E°, V
$\text{AgCl}(\text{s}) + e^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	0.222 V
$\text{Cu}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Cu}(\text{s})$	0.337 V

- (A) -0.115 V (B) -0.107 V
 (C) 0.107 V (D) 0.115 V

54. The standard cell potential of the following galvanic cell is 1.562 V at 298 K.



What is the cell potential of the following galvanic cell at 298 K?



- (A) 1.602 V (B) 1.626 V
 (C) 1.642 V (D) 1.691 V

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55. The mineral crocidolite has the formula $\text{Na}_2\text{Fe}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$. How many of the iron ions per formula unit are in the +2 oxidation state and how many are in the +3 oxidation state?

(A) All five are +2
(B) Three are +2, two are +3
(C) Two are +2, three are +3
(D) One is +2, four are +3

56. What products are formed in the electrolysis of 1.0 M aqueous HBr?

(A) H_2 at the cathode, Br_2 at the anode
(B) O_2 at the cathode, H_2 at the anode
(C) OH^- at the cathode, HOBr at the anode
(D) Br_3^- at the cathode, HBrO_4 at the anode

57. A 1.00 g sample of a hydrogen peroxide (H_2O_2) solution is placed in an Erlenmeyer flask and diluted with 20 mL of 1 M aqueous sulfuric acid. To this solution is added 0.0200 M KMnO_4 solution via a buret, until a pale purple color persists. This requires 22.50 mL of KMnO_4 solution. What is the percent by mass of hydrogen peroxide in the original solution?

(A) 0.613% (B) 1.53% (C) 3.83% (D) 7.65%

58. What is the standard cell potential of the following electrochemical cell?



Half-Reaction	E° , V
$\text{Ni}^{2+}(aq) + 2e^- \rightarrow \text{Ni}(s)$	-0.23
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80

(A) -1.83 V (B) 0.57 V (C) 1.03 V (D) 1.83 V

59. Copper electrodes are placed into two aqueous solutions of copper(II) sulfate at 25 °C. One compartment contains a 1.0 M solution while the other compartment contains a 0.10 M solution. The two compartments are connected with a salt bridge and the electrodes are connected by a wire passing through a voltmeter. In what direction do the electrons flow through the wire, and what is the cell potential read on the voltmeter?

Direction of electron flow	Cell potential
(A) From the electrode in the 1.0 M solution to the electrode in the 0.10 M solution	30 mV
(B) From the electrode in the 1.0 M solution to the electrode in the 0.10 M solution	59 mV
(C) From the electrode in the 0.10 M solution to the electrode in the 1.0 M solution	30 mV
(D) From the electrode in the 0.10 M solution to the electrode in the 1.0 M solution	59 mV

60. A 1.00 g sample of a silver-containing ore is dissolved in dilute nitric acid. The solution is neutralized and then selectively electrolyzed to deposit the silver metal, requiring 670 s of 0.10 A current. What is the mass percentage of silver in the ore?

(A) 2.5% (B) 3.0% (C) 3.7% (D) 7.5%

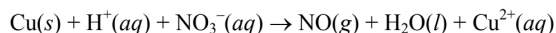
61. Which is formed at the cathode during the electrolysis of aqueous AgF ?

(A) $\text{Ag}(s)$ (B) $\text{H}_2(g)$ (C) $\text{O}_2(g)$ (D) $\text{F}_2(g)$

62. In which of the following substances is chlorine in the lowest oxidation state?

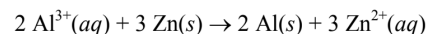
(A) Cl_2 (B) KCl (C) KClO (D) KClO_4

63. When the following reaction is balanced, what is the ratio of coefficients of $\text{H}^+(aq)$ to $\text{NO}(g)$?



(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 4 : 1

64. What is E° for the following reaction?



Half-Reaction	E° , V
$\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$	-1.66
$\text{Zn}^{2+}(aq) + 2e^- \rightarrow \text{Zn}(s)$	-0.76

(A) -1.04 V (B) -0.90 V
(C) 0.90 V (D) 1.04 V

Oxidation - Reduction

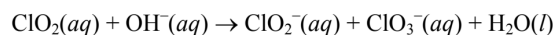
65. Electrolysis of 1.00 g of a copper(II) salt required passage of 0.100 A of current for 123 min for complete deposition of all the copper metal. What is the copper salt?
- (A) CuCl_2 , $M = 134.5$ (B) CuBr_2 , $M = 223.4$
 (C) $\text{Cu}(\text{NO}_3)_2$, $M = 187.6$ (D) $\text{Cu}(\text{ClO}_4)_2$, $M = 262.5$

66. What is the K_{sp} of Hg_2Cl_2 at 298 K?

Half-Reaction	E° , V (at 298 K)
$\text{Hg}_2^{2+} + 2e^- \rightarrow 2 \text{Hg}(l)$	+0.80
$\text{Hg}_2\text{Cl}_2(s) + 2e^- \rightarrow 2 \text{Hg}(l) + 2 \text{Cl}^-(aq)$	+0.31

- (A) 2.6×10^{-17} (B) 3.3×10^{-11}
 (C) 5.1×10^{-9} (D) 5.7×10^{-6}

67. When the equation



is balanced, what is the ratio of the coefficient of ClO_2 to that of ClO_3^- ?

- (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 3 : 2

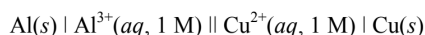
68. When an aqueous solution of KI is electrolyzed, what forms at the anode?

- (A) O_2 (B) I_2 (C) K (D) H_2O

69. A current of 0.15 A is passed through an aqueous solution of K_2PtCl_4 . How long will it take to deposit 1.00 g $\text{Pt}(s)$ ($M = 195.1$)?

- (A) 1600 s (B) 3300 s (C) 6600 s (D) 13000 s

70. In the galvanic cell

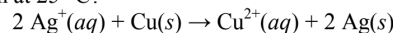


which of the following changes will increase the cell potential?

- I. Dilution of the Al^{3+} solution to 0.001 M
 II. Dilution of the Cu^{2+} solution to 0.001 M
 III. Increasing the surface area of the $\text{Al}(s)$ electrode

- (A) I only (B) II only
 (C) III only (D) I and III only

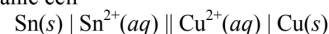
71. What is the equilibrium constant for the following reaction at 25 °C?



Half-Reaction	E° , V
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$	+0.34

- (A) 6.0×10^7 (B) 3.6×10^{15}
 (C) 3.6×10^{38} (D) 4.2×10^{42}

72. In the galvanic cell



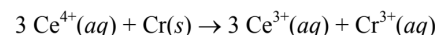
the standard potential is 0.48 V. Starting with standard concentrations, what are the concentrations of Sn^{2+} and Cu^{2+} when the cell has discharged to a potential of 0.45 V?

- (A) $[\text{Sn}^{2+}] = 0.47 \text{ M}$, $[\text{Cu}^{2+}] = 1.53 \text{ M}$
 (B) $[\text{Sn}^{2+}] = [\text{Cu}^{2+}] = 1.00 \text{ M}$
 (C) $[\text{Sn}^{2+}] = 1.53 \text{ M}$, $[\text{Cu}^{2+}] = 0.47 \text{ M}$
 (D) $[\text{Sn}^{2+}] = 1.82 \text{ M}$, $[\text{Cu}^{2+}] = 0.18 \text{ M}$

73. What is the oxidation number of C in formaldehyde, CH_2O ?

- (A) -2 (B) 0 (C) +2 (D) +4

74. In a galvanic cell in which the following spontaneous reaction takes place, what process occurs at the cathode?



- (A) Reduction of $\text{Cr}^{3+}(aq)$
 (B) Reduction of $\text{Ce}^{4+}(aq)$
 (C) Oxidation of $\text{Cr}(s)$
 (D) Oxidation of $\text{Ce}^{3+}(aq)$

75. Which two half reactions, when coupled, will make a galvanic cell that will produce the largest voltage under standard conditions?

- I. $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$ $E^\circ = +0.34 \text{ V}$
 II. $\text{Pb}^{2+}(aq) + 2e^- \rightarrow \text{Pb}(s)$ $E^\circ = -0.13 \text{ V}$
 III. $\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$ $E^\circ = +0.80 \text{ V}$
 IV. $\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$ $E^\circ = -1.66 \text{ V}$

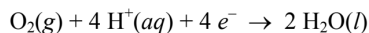
- (A) I and II (B) I and IV
 (C) II and IV (D) III and IV

76. An electrolysis cell is operated for 3000 s using a current of 1.50 A. From which 1.0 M solution will the greatest mass of metal be deposited?

- (A) TlNO_3 (B) $\text{Pb}(\text{NO}_3)_2$
 (C) ZnCl_2 (D) $\text{In}(\text{NO}_3)_3$

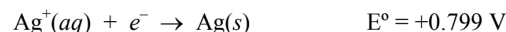
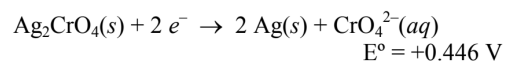
Oxidation - Reduction

77. The reduction of O_2 to H_2O in acidic solution has a standard reduction potential of +1.23 V. What is the effect on the half-cell potential at 25 °C when the pH of the solution is increased by one unit?



- (A) The half-cell potential decreases by 59 mV.
 (B) The half-cell potential increases by 59 mV.
 (C) The half-cell potential decreases by 236 mV.
 (D) The half-cell potential increases by 236 mV.

78. Given the two standard reduction potentials below, what is the K_{sp} of Ag_2CrO_4 at 25 °C?

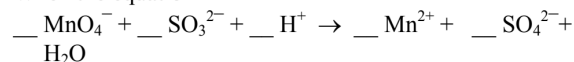


- (A) 8.64×10^{11} (B) 1.08×10^{-6}
 (C) 1.16×10^{-12} (D) 1.11×10^{-39}

79. What is the oxidation number of As in the compound $K(NH_4)_2AsO_4 \cdot 6H_2O$?

- (A) -3 (B) +1 (C) +3 (D) +5

80. When the equation



is balanced correctly with the smallest whole number coefficients, what is the coefficient for H_2O ?

- (A) 3 (B) 5 (C) 8 (D) 10

Half-Reaction	E° (V)
$2 H^+ + 2 e^- \rightarrow H_2$	0.00
$Pd^{2+} + 2 e^- \rightarrow Pd$	0.90
$O_2 + 4 H^+ + 4 e^- \rightarrow 2 H_2O$	1.23

According to the equations and data in the table above, which species is the strongest reducing agent of the following choices?

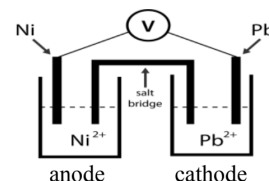
- (A) H^+ (B) H_2 (C) H_2O (D) Pd^{2+}

Half-Reaction	E° (V)
$Zn^{2+}(aq) + 2 e^- \rightarrow Zn(s)$	-0.76
$Cr^{3+}(aq) + 3 e^- \rightarrow Cr(s)$	-0.744
$Fe^{2+}(aq) + 2 e^- \rightarrow Fe(s)$	-0.409

Use the E° values in the table above to determine which of the following reactions will give the highest potential in a voltaic cell.

- (A) $3 Zn^{2+}(aq) + 2 Cr(s) \rightarrow 3 Zn(s) + 2 Cr^{3+}(aq)$
 (B) $3 Zn(s) + 2 Cr^{3+}(aq) \rightarrow 3 Zn^{2+}(aq) + 2 Cr(s)$
 (C) $Zn^{2+}(aq) + Fe(s) \rightarrow Zn(s) + Fe^{2+}(aq)$
 (D) $Zn(s) + Fe^{2+}(s) \rightarrow Zn^{2+}(aq) + Fe(s)$

83. The cell potential for the voltaic cell depicted below is 0.109 V under standard conditions, 1M $Ni^{2+}(aq)$ and 1 M $Pb^{2+}(aq)$.



Which change will increase the voltage?

- (A) The 1 M Ni^{2+} solution is diluted with H_2O .
 (B) A larger Ni electrode is used.
 (C) 50 mL of 1 M NaCl solution is added to precipitate $PbCl_2$.
 (D) More 1 M Pb^{2+} solution is added to that half-cell.

84. Aluminum is produced commercially by the electrolysis of Al_2O_3 . How many hours would be required to produce 250. g of Al using a 5.00 Ampere current?

- (A) 49.7 (B) 149 (C) 745 (D) 4020

85. Three metals, A, B and C, with solutions of their respective cations are tested in a voltaic cell with the following results;

A and B: A is the cathode

B and C: C is the cathode

A and C: A is the anode

What is the order of the reduction potentials from highest to lowest for the cations of these metals?

- (A) $A > B > C$ (B) $B > C > A$
 (C) $C > A > B$ (D) $B > A > C$

86. In which pair of substances do the nitrogen atoms have the same oxidation state?

- (A) HNO_3 and N_2O_5 (B) NO and HNO_2
 (C) N_2 and N_2O (D) HNO_2 and HNO_3

Oxidation - Reduction

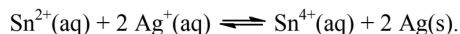
87. In the equation below, which species acts as the oxidizing agent? $\text{Pb(s)} + \text{PbO}_2\text{(s)} + 2 \text{H}^+\text{(aq)} + 2 \text{HSO}_4^-\text{(aq)} \rightarrow 2 \text{PbSO}_4\text{(s)} + 2 \text{H}_2\text{O(l)}$

(A) Pb(s) (B) $\text{PbO}_2\text{(s)}$
(C) $\text{H}^+\text{(aq)}$ (D) $\text{HSO}_4^-\text{(aq)}$

88. A standard voltaic cell is constructed using Cu metal in 1.0 M $\text{Cu(NO}_3)_2\text{(aq)}$ and an unknown metal in a 1.0 M solution of its nitrate salt. The cell voltage is 0.47 V when the Cu half-cell is the cathode. What is the standard reduction potential of the unknown metal? [$E^\circ_{\text{Cu}} = 0.34 \text{ V}$]

(A) -0.81 V (B) -0.13 V (C) 0.13 V (D) 0.81 V

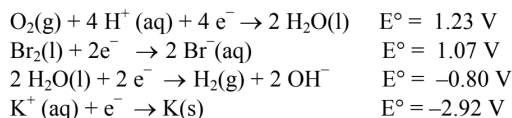
89. A voltaic cell is constructed with the overall reaction:



Which change will increase the voltage of the cell?

(A) increasing $[\text{Sn}^{2+}]$ (B) increasing $[\text{Sn}^{4+}]$
(C) decreasing $[\text{Ag}^+]$ (D) reducing the size of the Ag electrode

90. Use the standard reduction potentials to determine what is observed at the cathode during the electrolysis of a 1.0 M solution of KBr that contains phenolphthalein. What observation(s) is(are) made?



- (A) Solid metal forms.
(B) Bubbles form and a pink color appears.
(C) Dark red $\text{Br}_2\text{(aq)}$ forms.
(D) Bubbles form and the solution remains colorless.

91. Which species has an atom with an oxidation number of +3?

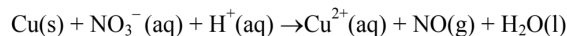
(A) ClO_2^- (B) PO_4^{3-} (C) $\text{S}_2\text{O}_3^{2-}$ (D) NO_2^+

92. According to the reduction potentials in the table below, which statement is true at standard conditions?

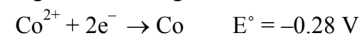
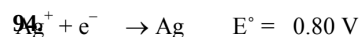
Reaction	E°, V
$\text{L}^{2+} + 2\text{e}^- \rightarrow \text{L}$	-0.13
$\text{M}^{2+} + 2\text{e}^- \rightarrow \text{M}$	-0.44
$\text{N}^{2+} + 2\text{e}^- \rightarrow \text{N}$	-0.76

- (A) L^{2+} ions oxidize M metal.
(B) M metal reduces N^{2+} ions.
(C) M is a better reducing agent than N.
(D) M^{2+} ions are better oxidizing agents than L^{2+} ions.

93. How many H^+ ions are required when the equation below is balanced with the smallest whole number coefficients?

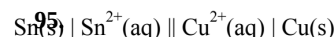


(A) 2 (B) 4 (C) 6 (D) 8



What is the E° value for the voltaic cell using the half-reactions above at standard conditions?

(A) 0.52 V (B) 1.08 V (C) 1.32 V (D) 1.88 V



For the voltaic cell represented above, which change will increase the voltage?

- (A) increasing the size of the Sn electrode
(B) increasing the size of the Cu electrode
(C) increasing the $[\text{Sn}^{2+}]$
(D) increasing the $[\text{Cu}^{2+}]$

96. For how many seconds must a current of 5.00 A flow in order to deposit 1.30 g of nickel from a solution of nickel(II) nitrate? (Coulombs = Amperes $\times$ seconds)

(A) 2.14×10^2 (B) 4.28×10^2
(C) 8.55×10^2 (D) 4.28×10^3

97. In which species does sulfur have the lowest oxidation state?

(A) SCl_2 (B) OSF_2 (C) H_2SO_3 (D) SF_6

98. All of the reactions below represent oxidation-reduction processes EXCEPT the

- (A) combustion of tin in chlorine gas.
(B) decomposition of potassium chlorate.
(C) neutralization of sodium hydroxide.
(D) reaction of magnesium with hydrochloric acid.

99. Which expression gives the correct value for the standard potential for a gold-rhodium voltaic cell?

Half-reaction	E°, V
$\text{Rh}^{3+}\text{(aq)} + 3\text{e}^- \rightarrow \text{Rh(s)}$	0.76
$\text{Au}^+\text{(aq)} + \text{e}^- \rightarrow \text{Au(s)}$	1.69

(A) $1.69 + 0.76$ (B) $1.69 - 0.76$
(C) $3(1.69) + 0.76$ (D) $3(1.69) - 0.76$

Oxidation - Reduction

100. When potassium permanganate, KMnO_4 , is added to an acidified solution of oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$, the products are CO_2 gas and Mn^{2+} ions. What is the reducing agent in this reaction?
- (A) KMnO_4 (B) $\text{H}_2\text{C}_2\text{O}_4$
(C) H_3O^+ (D) CO_2
101. Which term represents the charge on one mole of electrons?
- (A) one Ampere (B) one Coulomb
(C) one Faraday (D) one Volt
102. Consider a voltaic cell in which the reaction below occurs in two half-cells connected by a salt bridge and an external circuit.
 $2\text{Cr(s)} + 3\text{Sn}^{2+}(\text{aq}) \rightarrow 3\text{Sn(s)} + 2\text{Cr}^{3+}(\text{aq}) \quad E^\circ = 0.603 \text{ V}$
Which change will cause the voltage to increase?
- (A) Increasing the amount of Sn(s) in its half-cell
(B) Increasing the amount of Cr(s) in its half-cell
(C) Diluting the solution in the anode compartment
(D) Diluting the solution in the cathode compartment
103. How long must a current of 4.00 A be applied to a solution of $\text{Cu}^{2+}(\text{aq})$ to produce 2.0 grams of copper metal?
- (A) $2.4 \times 10^4 \text{ s}$ (B) $1.5 \times 10^3 \text{ s}$
(C) $7.6 \times 10^2 \text{ s}$ (D) $3.8 \times 10^2 \text{ s}$
104. What is the oxidation number of rhenium in $\text{Ca}(\text{ReO}_4)_2$?
- (A) +1 (B) +3 (C) +6 (D) +7
105. Consider the following reactions:
 $\text{X}(\text{NO}_3)_2 + \text{Y} \rightarrow \text{X} + \text{Y}(\text{NO}_3)_2$
 $\text{X}(\text{NO}_3)_2 + \text{Z} \rightarrow \text{X} + \text{Z}(\text{NO}_3)_2$
 $\text{Y}(\text{NO}_3)_2 + \text{Z} \rightarrow \text{No reaction}$
What is the correct order of increasing activity for the metals; X, Y, Z?
- (A) $\text{X} < \text{Y} < \text{Z}$ (B) $\text{X} < \text{Z} < \text{Y}$
(C) $\text{Z} < \text{Y} < \text{X}$ (D) $\text{Z} < \text{X} < \text{Y}$
106. For the cell
 $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g}) \quad E^\circ = 0.76 \text{ V}$
Which change will increase the voltage of the cell?
- (A) Increasing the size of the Zn electrode.
(B) Increasing the $[\text{Zn}^{2+}]$.
(C) Increasing the $[\text{H}^+]$.
(D) Increasing the pressure of the $\text{H}_2(\text{g})$.
107. When the half-reaction $\text{NO}_3^- \rightarrow \text{NO}$ is balanced for one NO_3^- in acid solution, _____ electron(s) is (are) _____.
(A) 3 gained (B) 1 gained
(C) 1 lost (D) 3 lost
108. In the reaction
 $\text{ClO}_3^-(\text{aq}) + 5\text{Cl}^-(\text{aq}) + 6\text{H}^+(\text{aq}) \rightarrow 3\text{Cl}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$
the oxidizing and reducing agents are, respectively,
(A) $\text{Cl}^-(\text{aq})$ and $\text{ClO}_3^-(\text{aq})$ (B) $\text{ClO}_3^-(\text{aq})$ and $\text{Cl}^-(\text{aq})$
(C) $\text{ClO}_3^-(\text{aq})$ and $\text{H}^+(\text{aq})$ (D) $\text{Cl}^-(\text{aq})$ and $\text{H}^+(\text{aq})$
109. $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag(s)} \quad E^\circ = 0.80 \text{ V}$
 $\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Co(s)} \quad E^\circ = -0.28 \text{ V}$
Use the standard reduction potentials to determine the standard potential for the reaction:
 $\text{Co(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Co}^{2+}(\text{aq}) + 2\text{Ag(s)}$
(A) 0.52V (B) 0.66V (C) 1.08V (D) 1.88V
110. What mass of copper is deposited when a current of 10.0A is passed through a solution of copper(II) nitrate for 30.6 seconds?
- (A) 0.101g (B) 0.201g (C) 0.403g (D) 6.04g
111. Which substance can act only as a reducing agent?
- (A) I_2 (B) BrCl (C) NaBr (D) HIO_4
112. When the equation
 $\text{Sn}^{2+}(\text{aq}) + \text{IO}_3^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + \text{I}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$
is balanced, what is the $\text{Sn}^{2+}(\text{aq}) / \text{IO}_3^-(\text{aq})$ mole ratio?
- (A) 1 / 1 (B) 2 / 1 (C) 1 / 2 (D) 5 / 2
113. Given the standard reduction potentials, which statement is correct?
- | |
|--|
| $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)} \quad E^\circ = 0.34 \text{ V}$ |
| $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) \quad E^\circ = 0.00 \text{ V}$ |
| $\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Cr(s)} \quad E^\circ = -0.73 \text{ V}$ |
- (A) Cr(s) will react with acid.
(B) Cu(s) will react with acid.
(C) $\text{Cu}^{2+}(\text{aq})$ will react with acid.
(D) Cu(s) will react with $\text{Cr}^{3+}(\text{aq})$.
114. Use the standard reduction potentials given to calculate the standard potential for the reaction;
 $\text{Pb(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Pb}^{2+}(\text{aq}) + 2\text{Ag(s)}$
- | |
|--|
| $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag(s)} \quad E^\circ = 0.80 \text{ V}$ |
| $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb(s)} \quad E^\circ = -0.13 \text{ V}$ |
- (A) 1.73 V (B) 0.93 V (C) 0.67V (D) 0.54 V

Oxidation - Reduction

115. For the voltaic cell represented,
 $\text{Ni(s)}|\text{Ni}^{2+}(\text{aq})||\text{Ag}^+(\text{aq})|\text{Ag(s)}$
 which change will increase the cell potential?
 (A) increasing the $[\text{Ag}^+]$ (B) increasing the $[\text{Ni}^{2+}]$
 (C) adding Ni(s) (D) removing Ag(s)
116. The deposition of 1.0 g of which element from its molten chloride requires the shortest time at a current of 1 A?
 (A) Na (B) Mg (C) Al (D) Ba
117. Which range includes the average oxidation state of S in $\text{Na}_2\text{S}_4\text{O}_6$?
 (A) less than 0 (B) 0 to +2
 (C) +2 to +4 (D) greater than +4
118. For a stoichiometric mixture of reactants, which statement best describes the changes that occur when this reaction goes to completion?
 $\text{Zn} + 4\text{HNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$
 (A) All of the zinc is oxidized and some of the nitrogen is reduced.
 (B) All of the zinc is oxidized and all of the nitrogen is reduced.
 (C) Some of the zinc is oxidized and all of the nitrogen is reduced.
 (D) Some of the zinc is oxidized and some of the nitrogen is reduced.
119. Which occurs at the anode of any voltaic cell?
 I. A metal electrode dissolves.
 II. A substance undergoes oxidation.
 III. Positive ions are deposited from the solution.
 (A) I only (B) II only
 (C) I and II only (D) I and III only
120. What is the coefficient for O_2 when the equation
 $\text{NH}_3 + \text{O}_2 \rightarrow \text{NO} + \text{H}_2\text{O}$
 is balanced with smallest whole number coefficients?
 (A) 2 (B) 3 (C) 4 (D) 5
121. What is the standard cell potential for the voltaic cell:
 $\text{Cr}|\text{Cr}^{3+}||\text{Pb}^{2+}|\text{Pb}$?

	$E^\circ_{\text{red}} / \text{V}$
$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$	-0.13
$\text{Cr}^{3+} + 3\text{e}^- \rightarrow \text{Cr}$	-0.74

 (A) 1.09 (B) 0.61 (C) -0.61 (D) -1.09
122. During the electrolysis of an aqueous solution of AgNO_3 , what would happen to the mass of silver metal deposited if the current is doubled and the electrolysis time is decreased to $\frac{1}{2}$ of its initial value?
 (A) It would stay the same.
 (B) It would increase to twice its initial value.
 (C) It would decrease to $\frac{1}{4}$ of its initial value.
 (D) It would decrease to $\frac{1}{2}$ of its initial value.
123. For the balanced equation:
 $8\text{H}^+(\text{aq}) + 5\text{Fe}^{2+}(\text{aq}) + \text{MnO}_4^-(\text{aq}) \rightarrow 5\text{Fe}^{3+}(\text{aq}) + \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
 which statement is correct?
 (A) $\text{Fe}^{2+}(\text{aq})$ undergoes oxidation
 (B) $\text{Fe}^{2+}(\text{aq})$ is the oxidizing agent
 (C) $\text{H}^+(\text{aq})$ undergoes oxidation
 (D) $\text{H}^+(\text{aq})$ is the oxidizing agent
124. Which species can act as an oxidizing agent but not as a reducing agent?
 (A) Cl_2 (B) Cl^- (C) ClO_2^- (D) ClO_4^-
125. What is the oxidation number of Ti in the compound $\text{Na}_2\text{Ti}_3\text{O}_7$?
 (A) -2 (B) +4 (C) +6 (D) +12
126. For a galvanic cell involving the half-reactions at standard conditions,
 $\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au} \quad E^\circ = 1.50 \text{ V}$
 $\text{Tl}^+ + \text{e}^- \rightarrow \text{Tl} \quad E^\circ = -0.34 \text{ V}$
 what is E°_{cell} ?
 (A) 0.48 V (B) 1.16 V (C) 1.84 V (D) 2.52 V
127. According to the half-reaction table,
 $\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn} \quad E^\circ = -0.14 \text{ V}$
 $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn} \quad E^\circ = -1.03 \text{ V}$
 which species is the better oxidizing agent?
 (A) Mn^{2+} (B) Sn^{2+} (C) Mn (D) Sn
128. The mass of metal deposited by the electrolysis of an aqueous solution of metal ions increases in direct proportion to which property?

I. electrolysis current
II. electrolysis time
III. metal ion charge

 (A) I only (B) III only
 (C) I and II only (D) I, II and III

Oxidation - Reduction

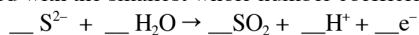
129. Which change represents an oxidation?

- (A) $\text{NO}_2^- \rightarrow \text{N}_2$ (B) $\text{VO}^{2+} \rightarrow \text{VO}_3^-$
 (C) $\text{ClO}^- \rightarrow \text{Cl}^-$ (D) $\text{CrO}_4^{2-} \rightarrow \text{Cr}_2\text{O}_7^{2-}$

130. What is the oxidation number of Mo in MoO_2Cl_2 ?

- (A) 0 (B) +3 (C) +5 (D) +6

131. What is the coefficient for H^+ when the half equation is balanced with the smallest whole number coefficients?



- (A) 2 (B) 4 (C) 6 (D) 8

132. What is the E° value for the voltaic cell constructed from the half-cells?

E° (V)	
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.762
$\text{Tl}^+ + \text{e}^- \rightarrow \text{Tl}$	-0.336

- (A) 0.090 V (B) 0.426 V
 (C) 1.098 V (D) 1.434 V

133. Which is a consistent set of values for a specific redox reaction carried out under standard conditions?

- | | E° | ΔG° | description |
|-----|-----------|------------------|----------------|
| (A) | + | - | spontaneous |
| (B) | - | + | spontaneous |
| (C) | + | + | nonspontaneous |
| (D) | - | - | nonspontaneous |

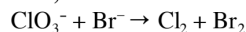
134. During the electrolysis of a dilute solution of sulfuric acid, what substance is produced at the anode?

- (A) hydrogen (B) hydrogen sulfide
 (C) oxygen (D) sulfur dioxide

135. Which represents an oxidation?

- (A) $\text{BrO}^- \rightarrow \text{Br}_2$ (B) $\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
 (C) $\text{Cr}^{3+} \rightarrow \text{CrO}_4^{2-}$ (D) $\text{VO}_3^- \rightarrow \text{VO}_2^+$

Questions 136 and 137 should be answered using the unbalanced equation;



136. Which is the reducing agent?

- (A) ClO_3^- (B) Br^- (C) Cl_2 (D) Br_2

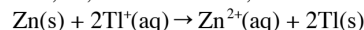
137. When this equation is balanced, what is the $\text{Br}^- / \text{ClO}_3^-$ ratio?

- (A) 1/1 (B) 2/1 (C) 3/1 (D) 5/1

Questions 138 and 139 require the use of these Standard Reduction Potentials.

Half Reaction	E° (V)
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.763
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$	-0.408
$\text{Tl}^+(\text{aq}) + \text{e}^- \rightarrow \text{Tl}(\text{s})$	-0.336
$\text{Cu}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Cu}^+(\text{aq})$	+0.161
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.769

138. Use the standard reduction potentials to find the standard cell potential, E° , for the reaction;



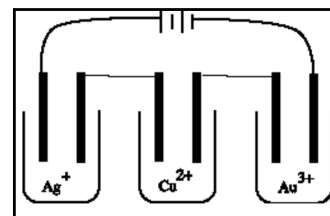
- (A) 0.427 V (B) 0.091 V
 (C) -0.091 V (D) -0.427 V

139. Based on the standard reduction potentials above, which reaction(s) is(are) spontaneous?

- I. $\text{Cr}^{2+}(\text{aq}) + \text{Fe}^{3+}(\text{aq}) \rightarrow \text{Cr}^{3+}(\text{aq}) + \text{Fe}^{2+}(\text{aq})$
 II. $\text{Cu}^{2+}(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \rightarrow \text{Cu}^+(\text{aq}) + \text{Fe}^{3+}(\text{aq})$

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

140. Solutions of AgNO_3 , CuSO_4 and AuCl_3 are electrolyzed in the apparatus depicted. If the electrolysis is stopped before any of the ions are deposited completely, how do the number of moles (n) of Ag, Cu and Au deposited compare?



- (A) $n_{\text{Ag}} = n_{\text{Cu}} = n_{\text{Au}}$ (B) $n_{\text{Ag}} < n_{\text{Cu}} < n_{\text{Au}}$
 (C) $n_{\text{Ag}} > n_{\text{Cu}} > n_{\text{Au}}$ (D) $n_{\text{Ag}} = n_{\text{Cu}} > n_{\text{Au}}$

141. In which species does the underlined element have an oxidation number of +2?

- (A) $\underline{\text{S}}\text{O}_2\text{Cl}_2$ (B) $\underline{\text{Fe}}(\text{CN})_6^{4-}$
 (C) $\underline{\text{H}}\text{NO}_2$ (D) $\underline{\text{Ni}}(\text{CO})_4$

142. Which transformation is an oxidation?

- (A) $\text{VO}_3^- \rightarrow \text{VO}_2^+$
 (B) $\text{CrO}_2^- \rightarrow \text{CrO}_4^{2-}$
 (C) $\text{SO}_3 \rightarrow \text{SO}_4^{2-}$
 (D) $\text{NO}_3^- \rightarrow \text{NO}_2^-$

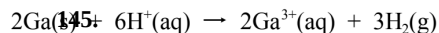
143. $__\text{Sn}^{2+}(\text{aq}) + __\text{NO}_3^-(\text{aq}) + __\text{H}^+(\text{aq}) \rightarrow __\text{Sn}^{4+}(\text{aq}) + __\text{NO}(\text{g}) + __\text{H}_2\text{O}$
 What is the coefficient for $\text{H}^+(\text{aq})$ when the equation above is balanced correctly with the smallest integer coefficients?

- (A) 2 (B) 4 (C) 6 (D) 8

Oxidation - Reduction

144. In electrochemical cells the cathode is always the electrode where

(A) oxidation occurs.
(B) reduction occurs.
(C) positive ions are formed.
(D) negative ions are formed.



The potential of the cell for the reaction given is 0.54 V. If the concentrations of the ions are 1.0 M and the pressure of $\text{H}_2(\text{g})$ is 1 atm, what is E° for the half-reaction $\text{Ga}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Ga}(s)$

(A) -0.54 V (B) -0.27 V
(C) 0.27 V (D) 0.54 V

146. All of the following affect the number of moles of metal deposited during electrolysis EXCEPT the

(A) current used (B) electrolysis time
(C) charge on the ion (D) molar mass

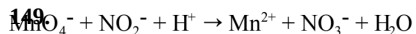
147. Which reaction occurs at the cathode during the electrolysis of an aqueous solution of KCl?

(A) $\text{K}^+(\text{aq}) + \text{e}^- \rightarrow \text{K}(s)$
(B) $2\text{H}_2\text{O}(l) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$
(C) $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
(D) $2\text{H}_2\text{O}(l) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$

148. Correct statements about a voltaic (galvanic) cell include which of the following?

1. Oxidation occurs at the anode.
2. Electrons flow from the cathode to the anode.

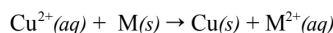
(A) 1 only (B) 2 only
(C) Both 1 and 2 (D) Neither 1 nor 2



When this equation is balanced correctly with the smallest integer coefficients, what is the coefficient for H^+ ?

(A) 1 (B) 6 (C) 8 (D) 16

150. An electrochemical cell constructed for the reaction:



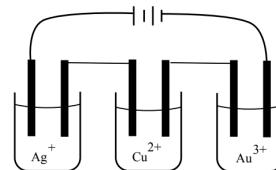
has an $E^\circ = 0.75\text{ V}$. The standard reduction potential for $\text{Cu}^{2+}(\text{aq})$ is 0.34 V. What is the standard reduction potential for $\text{M}^{2+}(\text{aq})$?

(A) 1.09 V (B) 0.410 V
(C) -0.410 V (D) -1.09 V

151. In which case does chromium undergo reduction?

(A) $\text{CrO}_3 \rightarrow \text{CrOF}_3$ (B) $\text{Cr}^{3+} \rightarrow \text{Cr}(\text{OH})_4^-$
(C) $2\text{CrO}_4^{2-} \rightarrow \text{Cr}_2\text{O}_7^{2-}$ (D) $\text{Cr}^{3+} \rightarrow \text{CrO}_4^{2-}$

152. 1.0 M aqueous solutions of AgNO_3 , $\text{Cu}(\text{NO}_3)_2$ and $\text{Au}(\text{NO}_3)_3$ are electrolyzed in the apparatus shown, so the same amount of electricity passes through each solution. If 0.10 moles of solid Cu are formed how many moles of Ag and Au are formed?



(A) 0.10 moles Ag, 0.10 moles Au
(B) 0.05 moles Ag, 0.075 moles Au
(C) 0.05 moles Ag, 0.15 moles Au
(D) 0.20 moles Ag, 0.067 moles Au

153. For which substance is the oxidation number of vanadium the same as that in the VO_3^- ion?

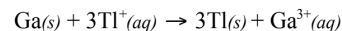
(A) VN (B) VCl_3
(C) VOSO_4 (D) VF_5



When this equation is balanced with whole number coefficients, what is the H^+/I_2 coefficient ratio?

(A) 2/1 (B) 3/1
(C) 6/1 (D) some other ratio

155. Use the information in the table to calculate E° for this reaction.



Reaction	E°
$\text{Ga}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Ga}(s)$	-0.529 V
$\text{Tl}^+(\text{aq}) + \text{e}^- \rightarrow \text{Tl}(s)$	-0.336 V

(A) 0.479 V (B) 0.193 V
(C) -0.193 V (D) -0.479 V

156. Nickel metal is added to a solution containing 1.0 M $\text{Pb}^{2+}(\text{aq})$ and 1.0 M $\text{Cd}^{2+}(\text{aq})$. Use the standard reduction potentials to determine which reaction(s) will occur.
Reaction 1 $\text{Ni}(s) + \text{Pb}^{2+}(\text{aq}) \rightarrow \text{Pb}(s) + \text{Ni}^{2+}(\text{aq})$
Reaction 2 $\text{Ni}(s) + \text{Cd}^{2+}(\text{aq}) \rightarrow \text{Cd}(s) + \text{Ni}^{2+}(\text{aq})$

Reaction	E°
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(s)$	-0.13 V
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(s)$	-0.23 V
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cd}(s)$	-0.40 V

(A) 1 only (B) 2 only
(C) both 1 and 2 (D) neither 1 nor 2

Oxidation - Reduction

157. What happens to a cation during the electrolysis of a molten salt? The cation moves toward the

- (A) anode and undergoes reduction.
- (B) anode and undergoes oxidation.
- (C) cathode and undergoes reduction.
- (D) cathode and undergoes oxidation.

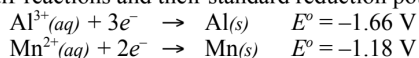
158. In which process does the nitrogen undergo oxidation?

- (A) $\text{N}_2 \rightarrow 2\text{NH}_3$
- (B) $\text{N}_2\text{O}_4 \rightarrow 2\text{NO}_2$
- (C) $2\text{NO}_3^- \rightarrow \text{N}_2\text{O}_5$
- (D) $\text{NO}_2^- \rightarrow \text{NO}_3^-$

159. Which statement is **not** true about the electrolysis of a 1 M solution of KI to which phenolphthalein has been added?

- (A) Potassium metal is formed.
- (B) A yellow color appears at the anode.
- (C) A pink color appears at the cathode.
- (D) A gas is produced at the cathode.

Questions 160 and 161 should be answered with reference to these half-reactions and their standard reduction potentials.

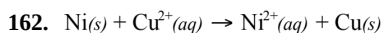


160. What process occurs at the anode of a voltaic cell utilizing these two half-reactions?

- (A) $\text{Al}(\text{s}) \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^-$
- (B) $\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$
- (C) $\text{Mn}(\text{s}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{e}^-$
- (D) $\text{Mn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mn}(\text{s})$

161. What is the standard potential of a voltaic cell produced by using these two half-reactions?

- (A) 0.04 V
- (B) 0.48 V
- (C) 2.84 V
- (D) 6.68 V



The voltaic cell based on this reaction has a voltage of 0.59 V under standard conditions. Which of these changes will produce a higher voltage?

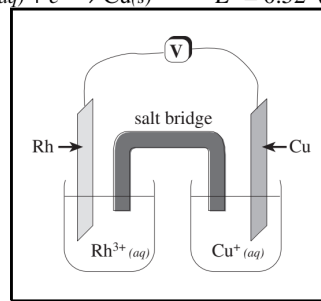
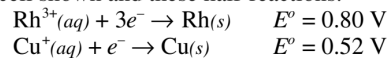
- I. increasing $[\text{Cu}^{2+}]$
- II. increasing the size of the $\text{Ni}(\text{s})$ electrode

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

163. Which equation represents an oxidation-reduction reaction?

- (A) $\text{H}_2\text{SO}_4 + 2\text{NH}_3 \rightarrow (\text{NH}_4)_2\text{SO}_4$
- (B) $\text{H}_2\text{SO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{CO}_2$
- (C) $2\text{K}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{Cr}_2\text{O}_7 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- (D) $2\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow \text{CuSO}_4 + 2\text{H}_2\text{O} + \text{SO}_2$

Questions 164 and 165 should be answered with reference to the voltaic cell shown and these half-reactions.



164. What is the direction of electron flow in the external circuit if the concentrations of Cu^+ and Rh^{3+} are each 1 M?

- (A) from the Rh anode to the Cu cathode
- (B) from the Rh cathode to the Cu anode
- (C) from the Cu anode to the Rh cathode
- (D) from the Cu cathode to the Rh anode

165. What is the voltage of this cell if the concentrations of Cu^+ and Rh^{3+} are each 1 M?

- (A) 0.28 V
- (B) 0.76 V
- (C) 1.32 V
- (D) 2.36 V

166. What is the correct order when the substances O_2 , H_2O , OF_2 , and H_2O_2 are arranged in order of increasing oxidation number for oxygen?

- (A) O_2 , H_2O , OF_2 , H_2O_2
- (B) H_2O , H_2O_2 , O_2 , OF_2
- (C) H_2O_2 , O_2 , H_2O , OF_2
- (D) OF_2 , O_2 , H_2O_2 , H_2O